

HDOC Day-31 — 2D Array Operations

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Algorithm

1. Start.
2. Read the number of rows and columns.
3. If rows or columns are not positive, display an error message and ask again.
4. Declare a 2D array using the given dimensions.
5. Read all matrix elements.
6. Repeatedly display the menu:
 - 1 — Sum of all matrix elements.
 - 2 — Find the sum of each row and store the sums in a 1D array.
 - 3 — Find and display the transpose of the matrix.
 - 4 — Exit.
7. Read the user's choice.
8. Use **switch**:
 - **Case 1:** Traverse the entire matrix, add all elements, and display the total.
 - **Case 2:** For every row, calculate its sum, store it in the row-sum array, and display the array.
 - **Case 3:** Display the matrix in transposed form by interchanging row and column indices.
 - **Case 4:** Exit.
 - **Default:** Display "Invalid choice!".
9. Repeat until choice 4 is selected.
10. Stop.

Test-Case Table

Test Case	Input	Operation	Expected Output
1	2×3 matrix: 1 2 3 / 4 5 6	Sum all	21
2	2×3 matrix: 1 2 3 / 4 5 6	Row sums	[6 , 15]
3	2×3 matrix: 1 2 3 / 4 5 6	Transpose	1 4 / 2 5 / 3 6
4	1×1 matrix: 5	Sum all	5
5	2×2 matrix: -1 2 / 3 -4	Sum all	0
6	Rows = 0	Validation	Invalid dimensions!

7	Menu choice = 7	Invalid menu	Invalid choice!
8	Choice = 4	Exit	Exiting program...

C Program

```
#include <stdio.h>

int main()
{
    int rows, cols, i, j, choice;

    /* Input and validation of dimensions */
    do
    {
        printf("Enter the number of rows and columns: ");
        scanf("%d %d", &rows, &cols);

        if (rows <= 0 || cols <= 0)
            printf("Invalid dimensions! Rows and columns must
be positive.\n");

    } while (rows <= 0 || cols <= 0);

    /* Variable Length Arrays */
    int matrix[rows][cols];
    int rowSum[rows];

    /* Input matrix */
    printf("\nEnter the matrix elements:\n");

    for (i = 0; i < rows; i++)
    {
        for (j = 0; j < cols; j++)
        {
            scanf("%d", &matrix[i][j]);
        }
    }

    do
    {
        printf("\n1. Find sum of all elements");
        printf("\n2. Find sum of each row");
        printf("\n3. Find transpose of matrix");
```

```

printf("\n4. Exit");

printf("\n\nEnter your choice: ");
scanf("%d", &choice);

switch (choice)
{
    case 1:
    {
        int total = 0;

        for (i = 0; i < rows; i++)
        {
            for (j = 0; j < cols; j++)
            {
                total = total + matrix[i][j];
            }
        }

        printf("Sum of all elements = %d\n", total);
        break;
    }

    case 2:
    {
        printf("\nRow sums:\n");

        for (i = 0; i < rows; i++)
        {
            rowSum[i] = 0;

            for (j = 0; j < cols; j++)
            {
                rowSum[i] = rowSum[i] + matrix[i][j];
            }

            printf("Sum of row %d = %d\n",
                i + 1, rowSum[i]);
        }

        printf("\nThe row-sum array contains: [");

        for (i = 0; i < rows; i++)
        {
            printf("%d", rowSum[i]);

```

```

        if (i < rows - 1)
            printf(", ");
    }

    printf("]\n");
    break;
}

case 3:
{
    printf("\nTranspose of the matrix:\n");

    for (j = 0; j < cols; j++)
    {
        for (i = 0; i < rows; i++)
        {
            printf("%d\t", matrix[i][j]);
        }

        printf("\n");
    }

    break;
}

case 4:
    printf("Exiting program...\n");
    break;

default:
    printf("Invalid choice! Please enter 1 to
4.\n");
}

} while (choice != 4);

return 0;
}

```

Execution

```
aaryanrajput@aaryans-macbook ~ % clang day31.c -o day31.c
aaryanrajput@aaryans-macbook ~ % ./day31.c
Enter the number of rows and columns: 2 3

Enter the matrix elements:
1 2 3 4 5 6

1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 1
Sum of all elements = 21

1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 2

Row sums:
Sum of row 1 = 6
Sum of row 2 = 15

The row-sum array contains: [6, 15]

1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 3

Transpose of the matrix:
1      4
2      5
3      6

1. Find sum of all elements
2. Find sum of each row
3. Find transpose of matrix
4. Exit

Enter your choice: 4
Exiting program...
aaryanrajput@aaryans-macbook ~ %
```